

Ayush Siddhant

✉ ayushsiddhant42@gmail.com

☎ +91 9661474996

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Backend Engineer | Distributed Systems, AI Infrastructure, Cloud Systems

Technical Skills

Backend & Languages: Python, TypeScript, JavaScript, C++, FastAPI, Node.js, WebSockets, Microservices

Databases: PostgreSQL, MongoDB, Redis, Weaviate, MySQL

DevOps & Cloud: Docker, Kubernetes, CI/CD, Prometheus, Grafana, AWS

AI/Agentic Systems: LangGraph, CrewAI, RAG, LLMs, LangChain, OpenAI Agents SDK

Experience

TheAgentic

Sep 2025 – Sept 2026

Backend Engineer — Full-time

Remote

- Built backend services using FastAPI and LangGraph to support automated workflows across multiple business use cases.
- Developed risk analysis pipeline by aggregating multi-source market data and generating automated token assessments.
- Built semantic search and matching system using embeddings and vector databases for candidate-job recommendations.

Qlaws.ai – Legal Solution

Jan 2025 – Aug 2025

Backend Engineer — Full-time

Remote

- Improved document processing pipeline using LangGraph orchestration and migrated 100K+ legal document vectors to Weaviate.
- Built document processing workflows handling extraction, validation, and generation tasks using specialized agents.
- Developed RAG-based backend services using Redis and PostgreSQL for document retrieval and conversational queries.

Dream Skrin – Supply Chain Intelligence

Oct 2024 – Jan 2025

Backend Engineer — Internship

Remote

- Designed backend intelligence layer connecting procurement, inventory, and logistics workflows.
- Built demand forecasting and automated reordering pipeline using XGBoost for supply chain optimization.
- Orchestrated distributed workflows using Temporal across multiple services.

MINTRIX – AI-Based Learning

May 2024 – Oct 2024

Backend Engineer — Internship

Remote

- Built RAG-powered chatbot using LangChain and Weaviate for student learning assistance.
- Improved API performance through caching and database query optimization.

Projects

GridLocalizer | Python, FastAPI, Redis Streams, PostgreSQL, React, TypeScript, Docker [🔗](#) [🐙](#)

- Built a control-room backend that turns noisy pole telemetry (live/dark only) into localized outage tickets at span, transformer, or feeder level, with one incident per real fault instead of per dark sensor.
- Designed ingest with Redis Streams and a worker for deduplication, debouncing, and graph-style localization; handled missing grid topology with inferred wiring and honest confidence scoring.
- Shipped operator UI, fault simulator, and telemetry-gated ticket closure; deployed publicly with Docker Compose and documented architecture, deployment, and system trade-offs.

KilnDB | Python, Binary Storage Engine, Write-Ahead Logging, B+ Trees, MVCC, Crash Recovery [🔗](#) [🐙](#)

- Building a relational database engine completely from scratch instead of using PostgreSQL or MySQL, implementing the storage layer, query execution, indexing, and recovery logic in Python.
- Designed a page-based storage engine with write-ahead logging (WAL), crash recovery, checksums, slotted pages, and fault injection tests to verify durability guarantees under simulated crashes.
- Implemented B+ Tree indexes, MVCC transactions, buffer pool management, and a SQL execution pipeline with a focus on understanding how production database internals work end to end.

PuneRental | Next.js, TypeScript, PostgreSQL, Supabase, MapLibre, Redis [🔗](#) [🐙](#)

- Built a map-first rental intelligence platform providing society-level rent insights, deposit trends, and bachelor accessibility signals for Pune.
- Designed PostgreSQL schemas and aggregation pipelines to process rental observations and compute median rent, price ranges, confidence metrics, and society intelligence scores.
- Developed Supabase-backed APIs with Redis caching, outlier detection, and community-driven trust scoring to improve data reliability and scalability.

Education

Symbiosis Institute of Technology, Pune

2027

B.Tech in Computer Science & Engineering